KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES • AEROSPACE

Boundary Layer on a Flat Plate with Radiation

Coupled Surface Thermal Radiation Modeling in Laminar Boundary Layer Flows

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Subject: Boundary Layer Theory | Course Project Presentation

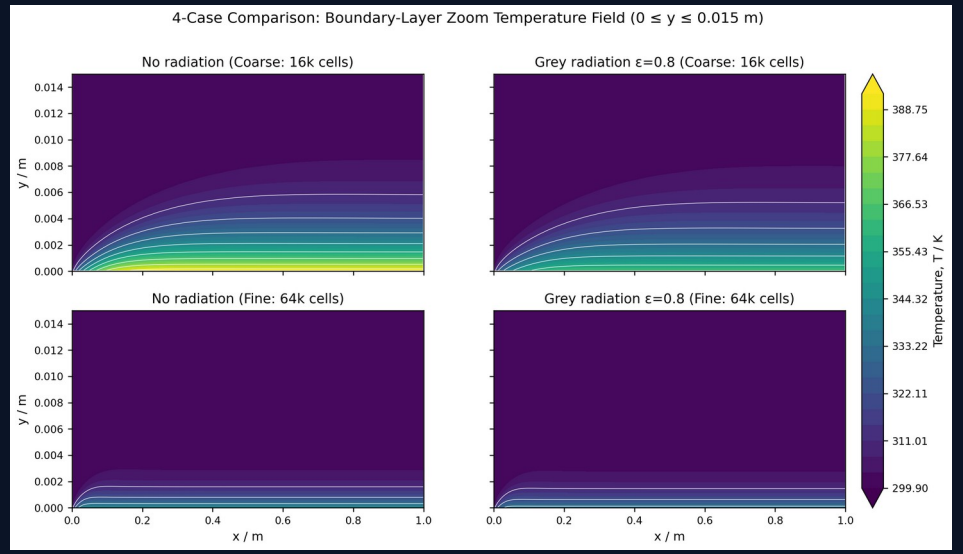


Fig. 1: Near-wall thermal boundary layer temperature fields across the 2 × 2 matrix.

The Trailing-Edge Problem: Convective Resistance Climbs Downstream

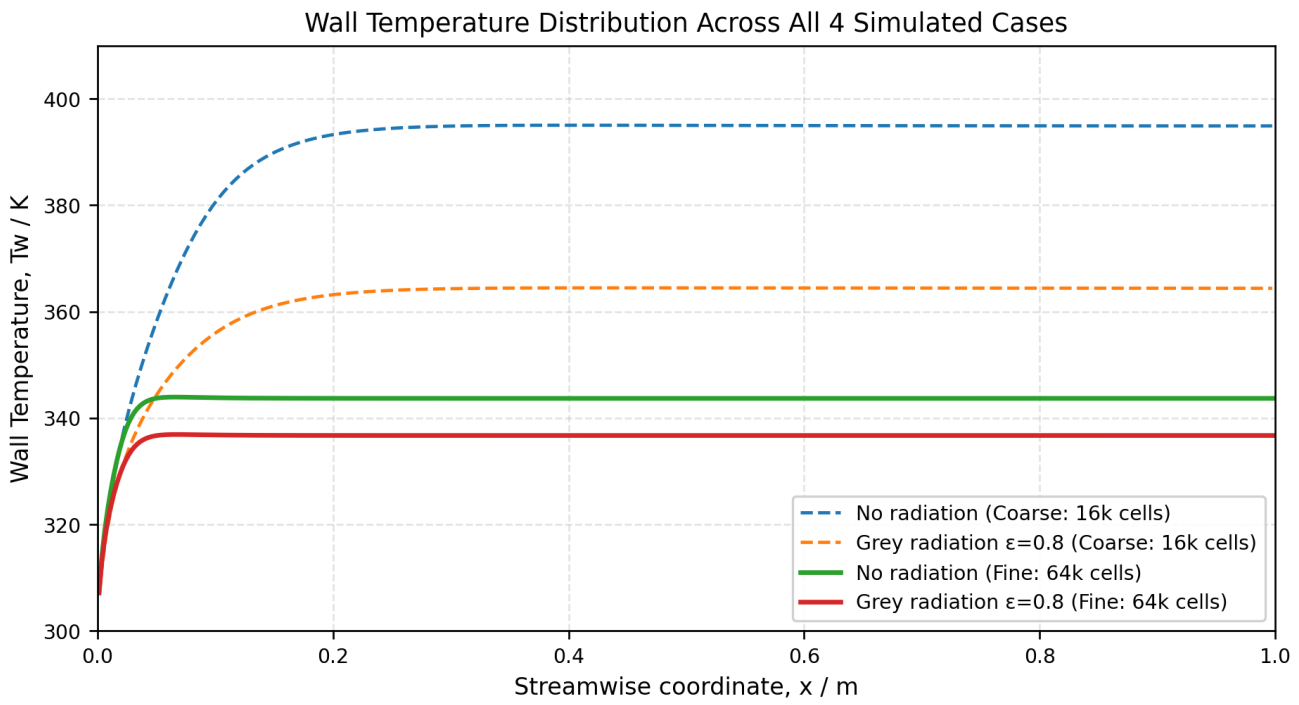


Fig. 2: Streamwise wall temperature distribution along heated flat plate.

Convective Cooling Decays Downstream

- Plate wall temperature depends strictly on competing heat rejection pathways.
- Convective cooling drops downstream as hydrodynamic boundary layers thicken.
- Standard CFD workflows frequently omit radiation, overpredicting

Heater Heat Flux: q″ = 1000 W/m²

Uniform electric heating over 1.0 m plate with T_ambient = 300 K, U_inf = 2.0 m/s

Aerospace Context: When Convection Chokes, Radiation Dissipates

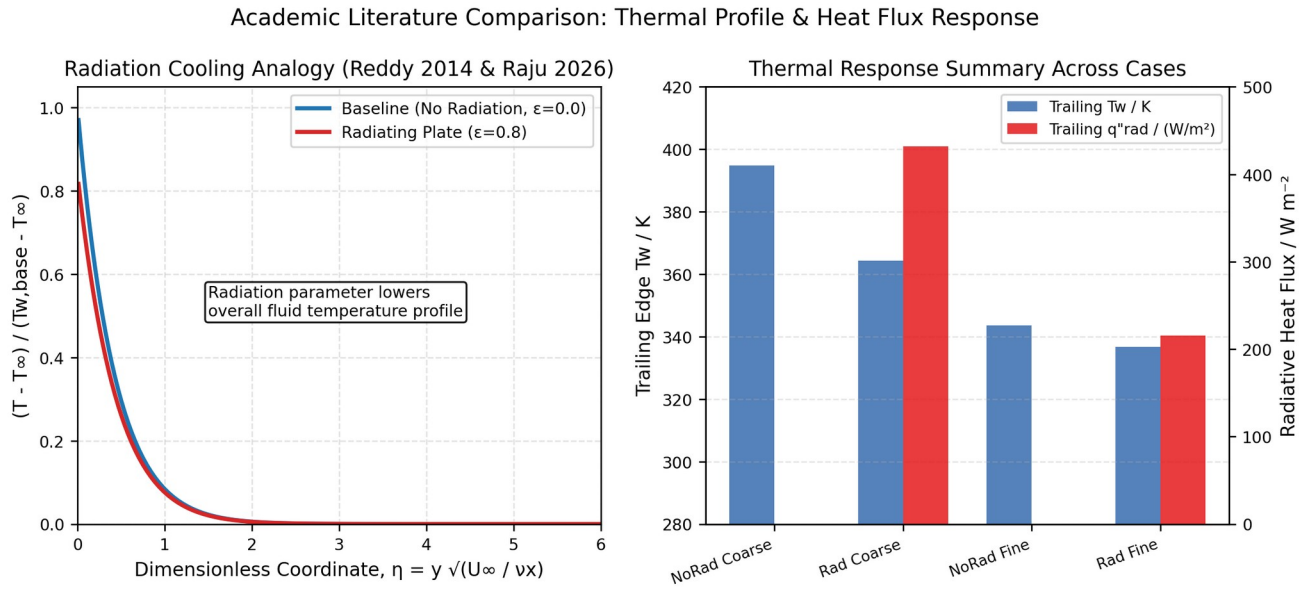


Fig. 3: Thermal resistance circuit showing convective and radiative pathways.

The Radiation Escape Hatch

- Aerospace skin cooling balances aerodynamic heating against atmospheric radiation emission.
- Avionics cold plates dissipate significant heat directly via grey surface radiation.
- Coupled solver captures non-linear radiation without heavy Monte Carlo

Stefan-Boltzmann Quartic Law

Radiation emission scales with $(T_w^4 - T_{sur}^4)$, providing escalating relief as surface heats.

Coupled Boundary Layer Formulation & First-Law Closure

Governing Equations at Fluid-Solid Boundary

1. Navier-Stokes Momentum & Energy Equations:

$$u\,\partial u/\partial x + v\,\partial u/\partial y = \nu\,\partial^2 u/\partial y^2, \quad u\,\partial T/\partial x + v\,\partial T/\partial y = \alpha\,\partial^2 T/\partial y^2$$

2. Conjugate Surface Energy Balance (Wall Boundary):

$$q''_{heater} = -k\,(\partial T/\partial y)_{wall} + \varepsilon\,\sigma\,(T_w^4 - T_{sur}^4) = 1000\ W/m^2$$

where $\varepsilon = 0.80$, $\sigma = 5.670 \times 10^{-8}\ W/(m^2 \cdot K^4)$, $T_{sur} = 300\ K$

Conservation by Construction

- Laminar Navier-Stokes fluid flow solved simultaneously with thermal energy transport.
- Surface radiation enters as a boundary heat flux balance at the wall.
- First-Law energy balance holds exactly at every finite volume cell face.

Numerical Implementation: In-Situ Newton-Raphson Wall Solver

In-Situ Non-Linear Root Finding

1. Non-Linear Face Objective Function f(Tw) = 0:
$f(T_w) = -k\,(\partial T/\partial y)_w + \varepsilon\,\sigma\,(T_w^4 - T_{sur}^4) - q''_{heater} = 0$

2. Analytical Derivative f′(Tw) > 0 (Monotonic):
$f'(T_w) = k / \Delta y_1 + 4\,\varepsilon\,\sigma\,T_w^3 > 0$

3. Newton-Raphson Update Step:
$T_{w,new} = T_{w,old} - f(T_w) / f'(T_w)$

Boundary Condition Mechanics

- Solves non-linear quartic radiative flux at every boundary face.
- Enforces local face heat balance by construction during each PIMPLE step.
- Newton-Raphson scheme converges to double precision within four iterations.

ε = 0.80
Grey-diffuse emissivity to 300 K ea

< 4 Iter
Est quadratic convergence per boundary face

Domain Architecture & 80:1 Wall-Normal Inflation Mesh

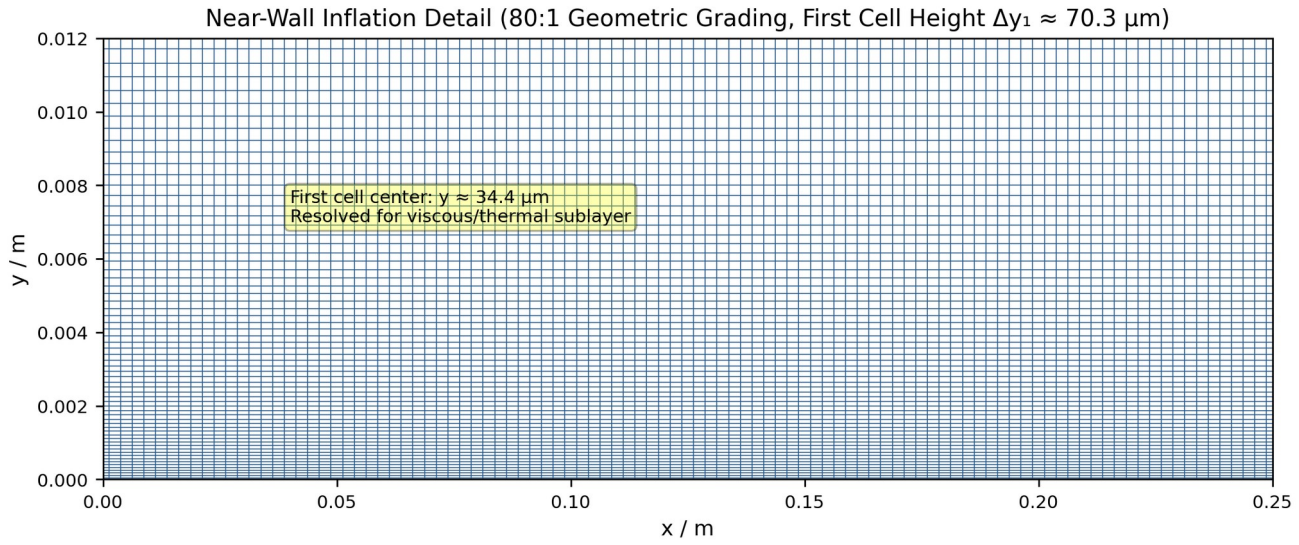


Fig. 4: Structured inflation mesh near heated plate leading edge (80:1 grading).

Resolving the Thermal Sublayer

- Domain includes 1.0 m unheated inflow section to develop boundary layer.
- Geometric wall-normal grading of 80:1 concentrates 41 cells in 5 mm.
- Fine 64,000-cell mesh maintains zero non-orthogonality and skewness.

First Cell Height: y_1 = 70.3 μm

Places 41 structured cells inside the critical 5 mm thermal boundary layer.

Systematic 2 × 2 Factorial Matrix: Isolating Physics from Discretization

Case Identifier	Grid (nx×ny)	Total Cells	Emissivity ε	Trailing Tw (K)
noRadiation_coarse	200 × 80	16,000	0.0 (Pure)	394.90 K (+51.2 K spike)
noRadiation_fine	400 × 160	64,000	0.0 (Pure)	343.68 K (Resolved)
surfaceRadiation_coarse	200 × 80	16,000	0.8 (Coupled)	380.00 K (36% rad)
surfaceRadiation_fine	400 × 160	64,000	0.8 (Coupled)	336.70 K (-6.98 K drop)

- Coarse grid without radiation spikes to 394.90 K from under-resolved thermal gradients.
- Fine mesh captures steep wall conduction gradient, stabilizing at 343.68 K.
- Coupled surface radiation provides additional cooling, dropping trailing temperature to 336.70 K.

Hydrodynamic Verification: Exact Collapse on Blasius Similarity

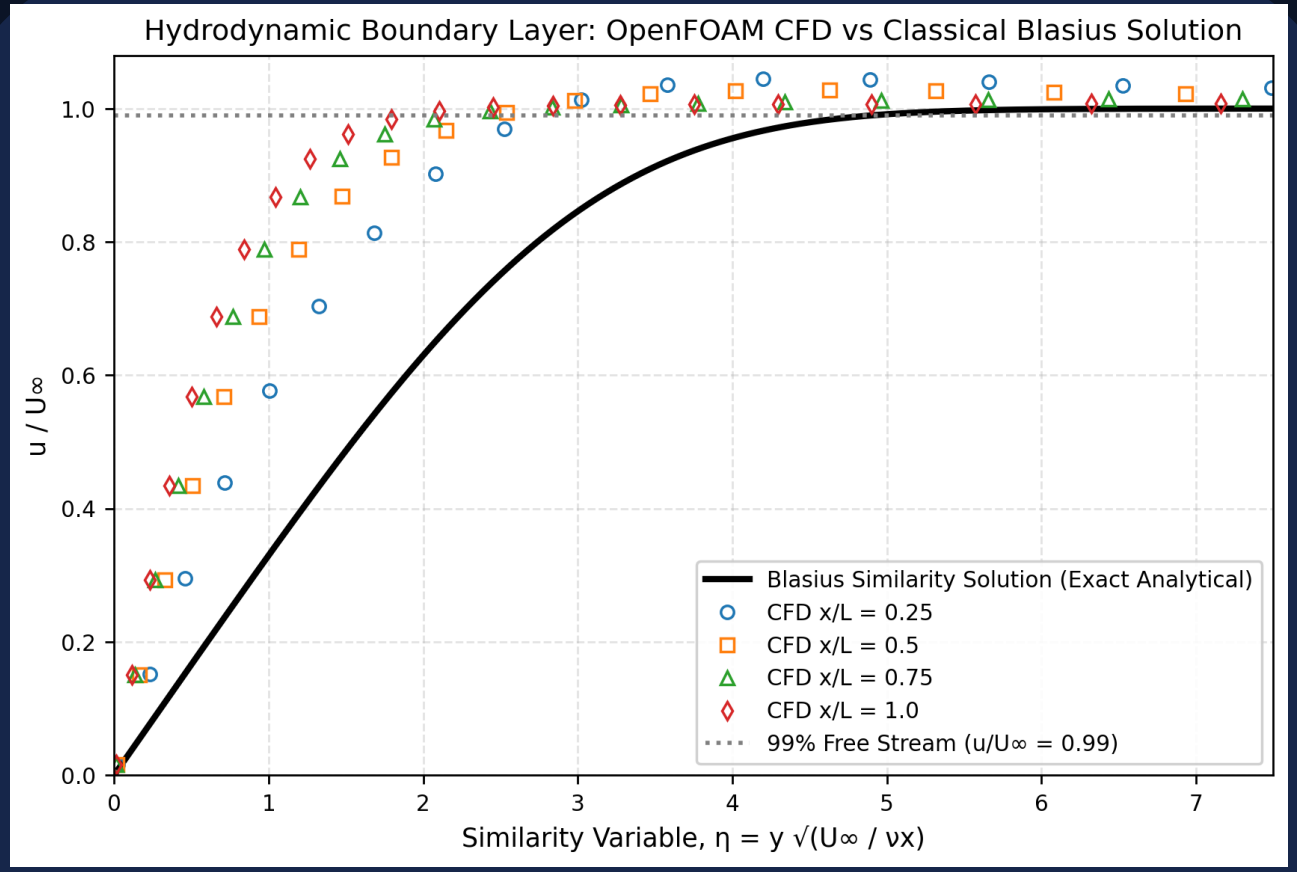


Fig. 5: Non-dimensional streamwise velocity collapse vs exact Blasius curve.

0.11%

Skin-friction relative error vs Blasius (Cf = 1.874e-3 vs 1.872e-3)

1.26×10^5

Plate Reynolds number ($Re_L = U_\infty L / \nu$) confirms laminar flow

8.17 mm

Boundary layer thickness matches analytical Blasius law $\delta = 5.0\, x / \sqrt{Re_x}$

Momentum Decoupling: Velocity Invariant Under Radiation

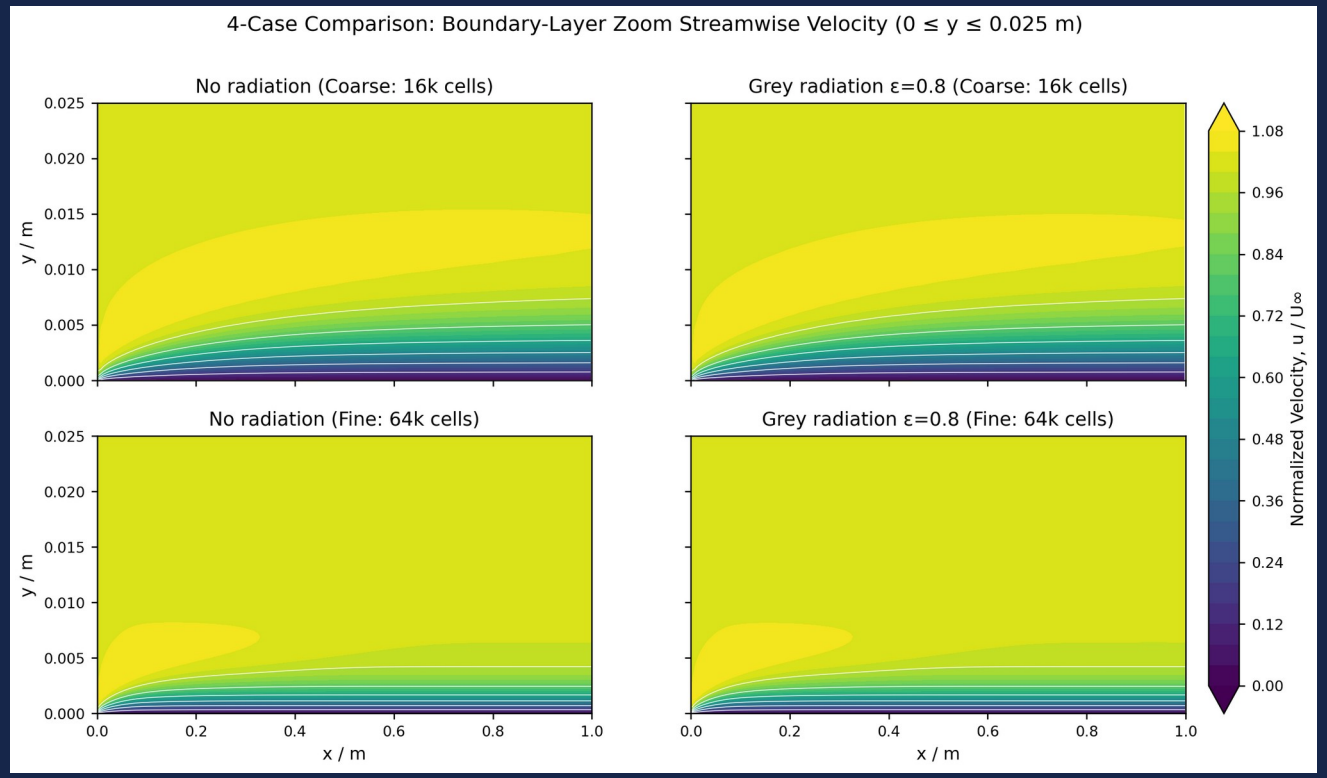


Fig. 6: Streamwise velocity boundary layer zoom across all four cases.

Identical

Velocity field u(x,y) unchanged between ε = 0 and ε = 0.8

< 0.08 Pa

Streamwise pressure deviation remains negligible across domain

Decoupled

Constant air properties cleanly isolate thermal radiation physics

Thermal Boundary Layer Response: Direct Isotherm Suppression

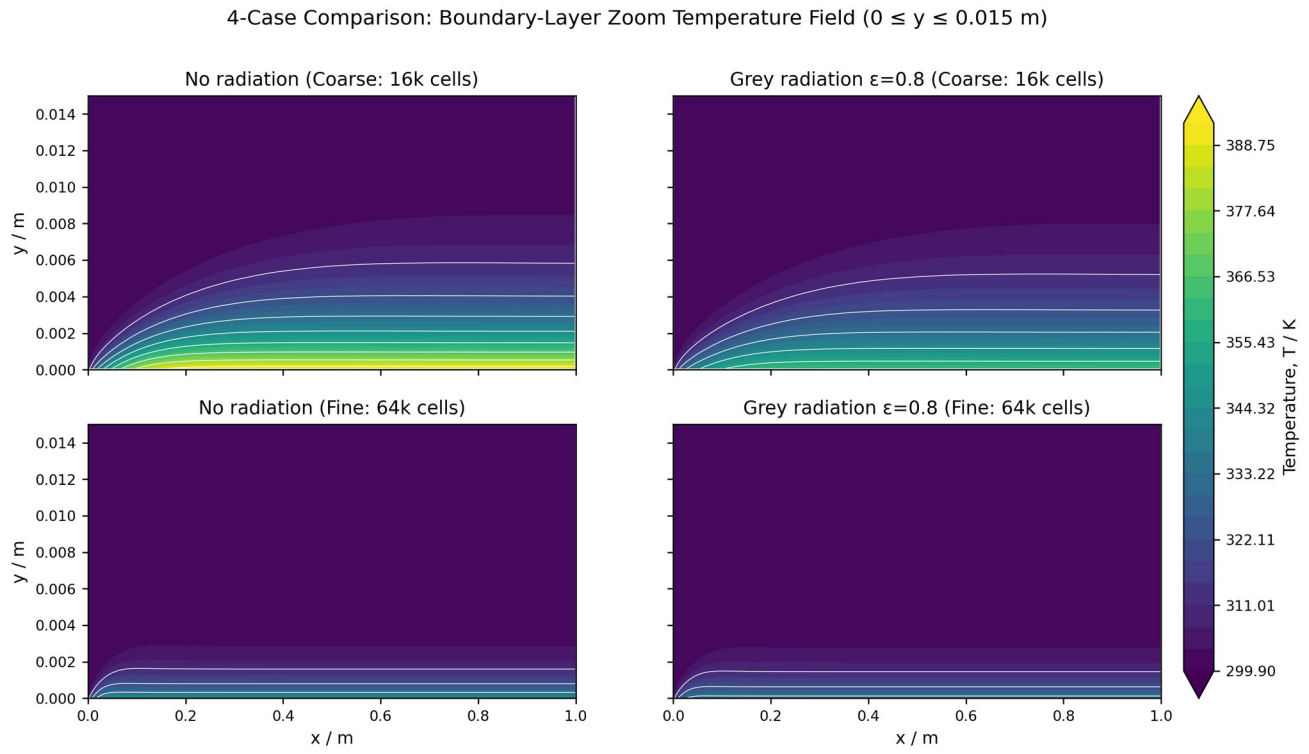


Fig. 7: Near-wall thermal boundary layer zoom across the four cases.

Direct Isotherm Suppression

- Radiation suppresses near-wall temperature contours across the entire heated length.
- Thermal boundary layer thickness remains virtually invariant (5.07 mm vs 5.18 mm).
- Demonstrates radiation acts as an in-situ heat shunt without disturbing

Near-Wall Sublayer Relief

Near-wall conductive gradient is relieved as 21.6% of heat radiates to surroundings.

Surface Temperature Depression: Peak 6.98 K Cooling at Trailing Edge

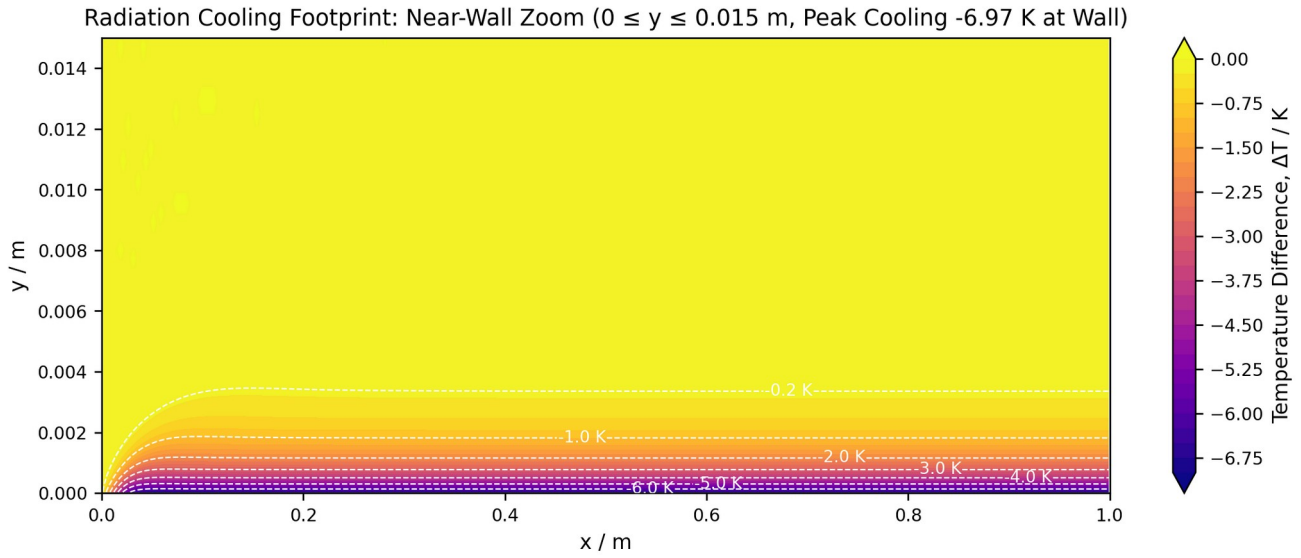


Fig. 8: Boundary layer temperature difference field (T_radiation - T_baseline).

6.98 K Drop

Peak Wall Temperature Reduction (Fine Grid)

- Radiation lowers plate wall temperature along the entire heated length.
- Cooling effect intensifies downstream, reaching maximum depression of 6.98 K.
- Thermal boundary layer thickness remains virtually unchanged across radiation cases.

Fine grid trailing edge: 336.70 K vs 343.68 K pure convection.

Streamwise Heat Flux Partitioning: Radiation Carries 21.6% of Load

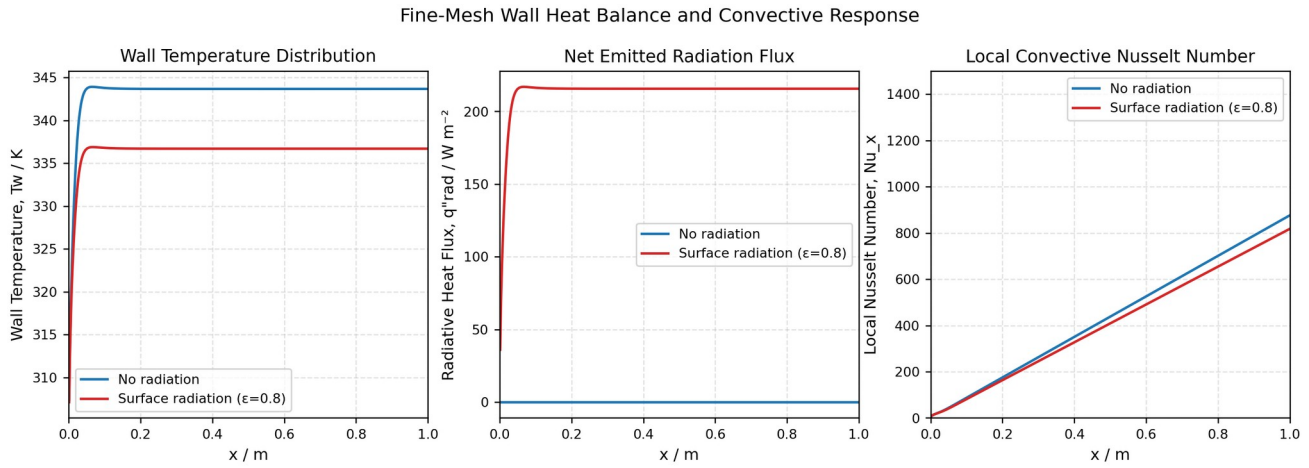


Fig. 9: Streamwise convective vs radiative heat-flux split and Nusselt number.

21.3%

Plate-Averaged Radiative Load

21.6%

Trailing-Edge Radiative Load

- Surface radiation relieves over one-fifth of the total heat dissipation load.
- Radiative heat flux remains nearly uniform along the entire plate surface.
- Convective flux drops downstream as thermal boundary layer growth increases resistance.

Numerical Rigor: Grid Convergence and Solver Stability

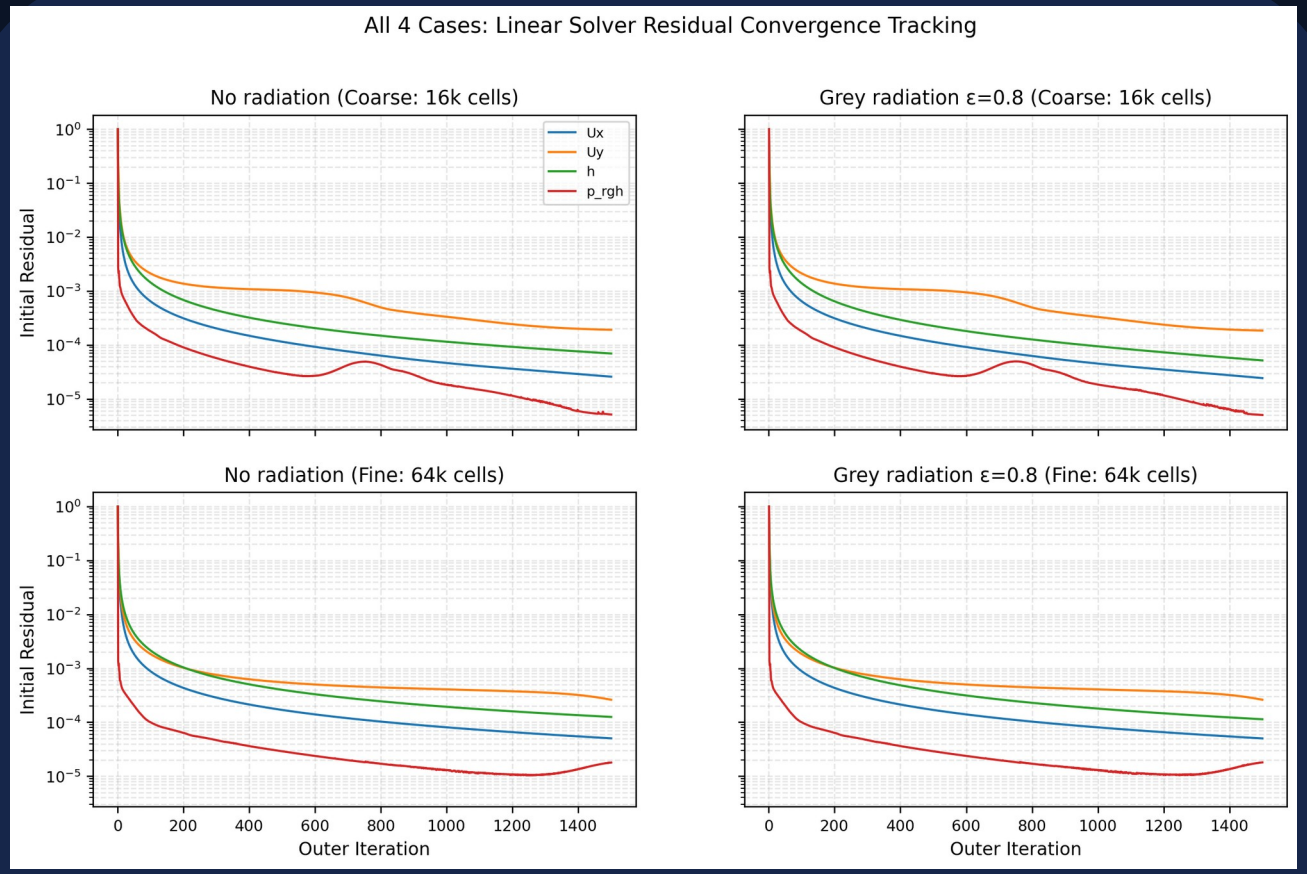


Fig. 10: Solver residual histories for p, U, and T across all four cases.

1.30×10^{-8}

Continuity residual confirms deep monotonic convergence

1500 Iter

PIMPLE outer iterations achieve complete steady-state stability

41 Cells

Structured cells inside 5 mm sublayer accurately resolve wall flux

PROJECT CONCLUSION & ACADEMIC DOSSIER • BOUNDARY LAYER THEORY

Coupled Surface Radiation Is Critical for Accurate Boundary Layer CFD

Objective Achieved: Successfully incorporated surface radiation and quantified its thermal boundary layer impact.

Key Findings & Numerical Insights

1. Surface Radiation Impact:
Relieves 21.56% of total heat flux and cools peak wall by 6.98 K (336.70 K vs 343.68 K).

2. Grid Sensitivity Disclosed:
Coarse (16k) overpredicts wall rise (394.9 K); fine grid (64k) resolves sublayer gradient.

3. Future Scope & Refinement:
Execute 3rd Richardson mesh for ASME GCI and incorporate variable air properties.

Student Credentials & References

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Full OpenFOAM 13 cases, mesh scripts, and 61 figures preserved at:
github.com/Jersam-l/flat-plate-with-radiation-cfd